

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:		Reg. No.:	
Date:		Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15 %)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15 %)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

CO5_AT1

**AI-Based Smart Renewable Energy Asset Performance
Monitoring System**

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DEPT: **BACHELOR OF TECHNOLOGY IN AIDS**

COURSE: **CSA1017 SOFTWARE ENGINEERING**

Research Paper Review / Literature Survey Report

AI-Based Smart Renewable Energy Asset Performance Monitoring System

1. Introduction

Renewable energy plants such as solar photovoltaic (PV) farms and wind farms require continuous monitoring to maintain high energy production and equipment reliability. Traditional periodic inspection methods may identify failures only after performance has already degraded. Recent research increasingly uses **Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), IoT, SCADA data, cloud computing, computer vision, and predictive analytics** for condition monitoring and predictive maintenance.

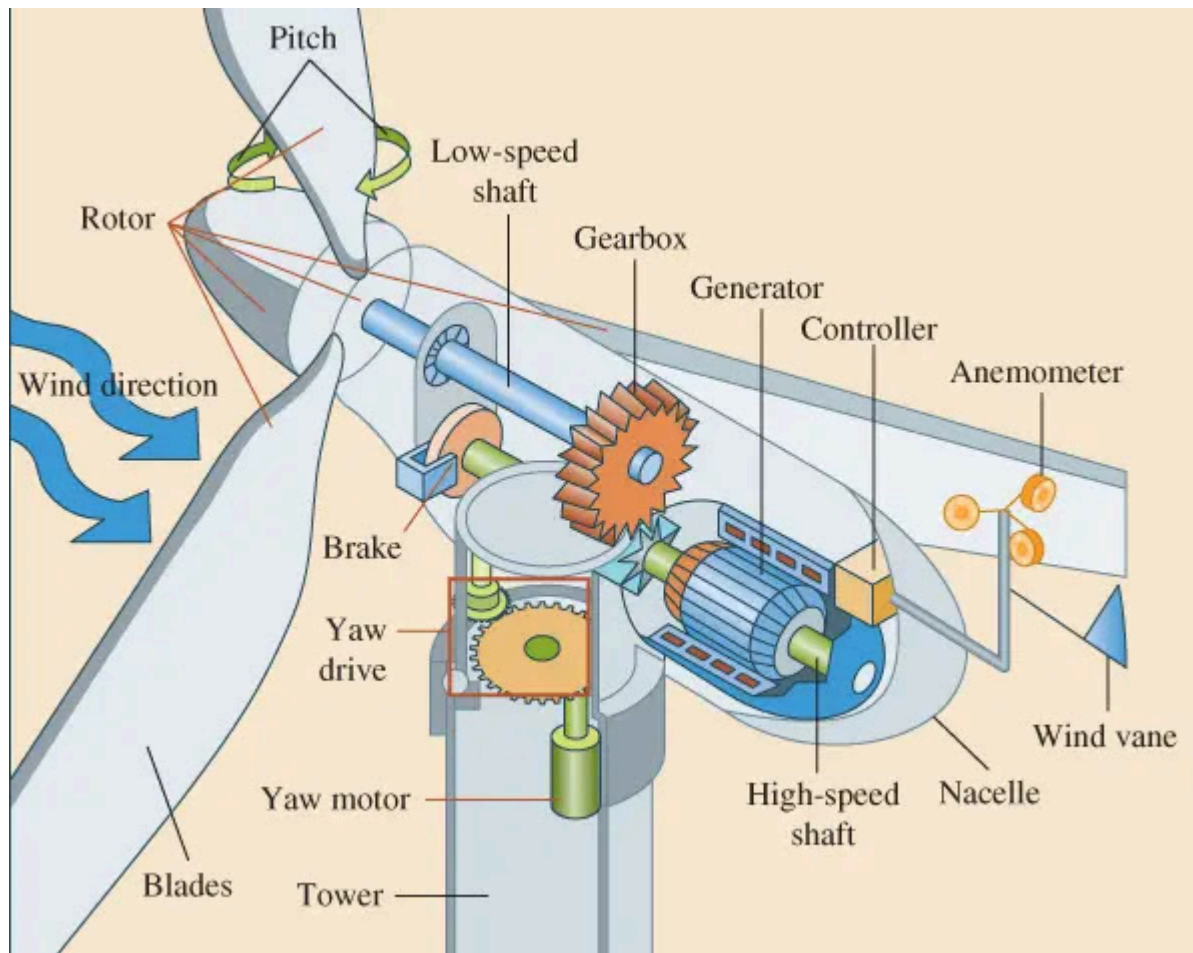
The literature reviewed below focuses mainly on four areas relevant to the proposed system: **wind-turbine fault detection, solar-PV fault detection, renewable-energy forecasting, and IoT/edge-based monitoring**. The studies demonstrate that AI can improve fault detection and energy prediction, but challenges remain in data availability, model generalization, explainability, real-time deployment, and integration across different renewable assets.

2. Selected Research Papers

The following papers were selected because they directly address AI-based monitoring, fault detection, predictive maintenance, and renewable-energy forecasting:

1. Chen et al. (2019) – Deep learning for imbalanced wind-turbine SCADA fault detection.
2. Xiang et al. (2021) – CNN-LSTM with attention for wind-turbine fault detection.
3. Khan et al. (2023) – Genetic algorithm-based ensemble learning for wind-turbine fault detection.
4. Duranay (2023) – Deep learning for solar-panel fault detection.
5. PV remote-monitoring study (2023) – IoT and machine learning for PV-array monitoring and diagnosis.
6. Maldonado-Correa et al. (2024) – Transformer-based wind-turbine fault detection.
7. Wind SCADA forecasting study (2022) – ML-based wind-power forecasting.
8. 2025/2026 research – Generative transfer learning for wind turbines with limited data.

These studies collectively provide a strong foundation for the proposed **AI-Based Smart Renewable Energy Asset Performance Monitoring System**.



3. Individual Research Paper Reviews

Paper 1: Learning Deep Representation of Imbalanced SCADA Data for Fault Detection of Wind Turbines

Authors/Year: Longting Chen, Guanghua Xu, Qing Zhang, Xun Zhang – 2019

Research Problem

Wind-turbine SCADA datasets contain a large number of normal operating samples but relatively few fault samples. This class imbalance can negatively affect machine-learning-based fault diagnosis.

Methodology

The researchers proposed a deep-learning approach capable of learning representations from imbalanced SCADA data. The method considers both local and global characteristics of the data.

Dataset/Technology

- Wind-turbine SCADA data
- Deep neural networks
- Feature representation learning
- Fault classification

Major Findings

The study demonstrated that deep representation learning can improve fault-detection performance and generalization when dealing with imbalanced wind-turbine data. ([ScienceDirect](#))

Limitation

The approach is primarily focused on wind turbines and does not provide a unified monitoring architecture for solar, wind, and battery assets.

Relevance

This research supports the proposed system's requirement for handling **imbalanced fault data**.

Paper 2: Fault Detection of Wind Turbine Based on SCADA Data Analysis Using CNN and LSTM with Attention Mechanism

Authors/Year: Ling Xiang, Penghe Wang, Xin Yang, Aijun Hu, Hao Su – 2021

Research Problem

Wind turbines operate under changing environmental and operational conditions, making early fault detection difficult.

Methodology

The researchers combined:

- CNN
- LSTM
- Attention Mechanism
- SCADA data
- EWMA-based anomaly detection

CNN extracts important features, while LSTM learns temporal patterns. The attention mechanism gives greater importance to relevant information. ([ScienceDirect](#))

Dataset/Technology

- Wind-turbine SCADA data
- CNN
- LSTM
- Attention mechanism
- EWMA

Major Findings

The model was able to identify abnormal operating conditions and provide early warnings of wind-turbine failures. ([ScienceDirect](#))

Limitation

Deep-learning models require considerable training data and computational resources. The study also focuses mainly on wind-turbine monitoring.

Relevance

The CNN-LSTM approach can be adapted to the proposed platform for **time-series equipment monitoring and failure prediction**.

Paper 3: SCADA System Dataset Exploration and Machine Learning Based Forecast for Wind Turbines

Year: 2022

Research Problem

Wind-power generation varies according to wind speed, direction, weather, and turbine operating conditions. Accurate forecasting is therefore necessary for efficient renewable-energy management.

Methodology

The study compared several machine-learning models:

- Random Forest
- XGBoost
- LSTM

The models used historical SCADA data and operational/weather-related variables. ([ScienceDirect](#))

Dataset

The study used wind-farm SCADA data containing variables such as:

- Wind speed
- Wind direction

- Theoretical power
- Active power
- Month
- Hour

Major Findings

XGBoost achieved approximately $R^2 = 0.969$, with reported MSE, RMSE, MAPE, and MAE values showing strong forecasting performance. ([ScienceDirect](#))

Limitation

The work concentrates on wind-power forecasting rather than complete asset-health monitoring.

Relevance

The forecasting component can help the proposed system identify differences between **expected and actual energy production**.

Paper 4: Fault Detection of Wind Turbines Using SCADA Data and Genetic Algorithm-Based Ensemble Learning

Authors/Year: Prince Waqas Khan, Chan Yeob Yeun, Yung Cheol Byun – 2023

Research Problem

Wind-turbine faults increase maintenance costs and downtime. Selecting the most useful SCADA variables is also challenging.

Methodology

The researchers developed an ensemble model combining:

- XGBoost
- Random Forest
- Extra Trees
- Genetic Algorithm

The genetic algorithm was used for feature selection, while the ensemble model was used for anomaly detection. ([ScienceDirect](#))

Dataset/Technology

- Wind-turbine SCADA data
- Genetic algorithm
- XGBoost
- Random Forest
- Extra Trees

Major Findings

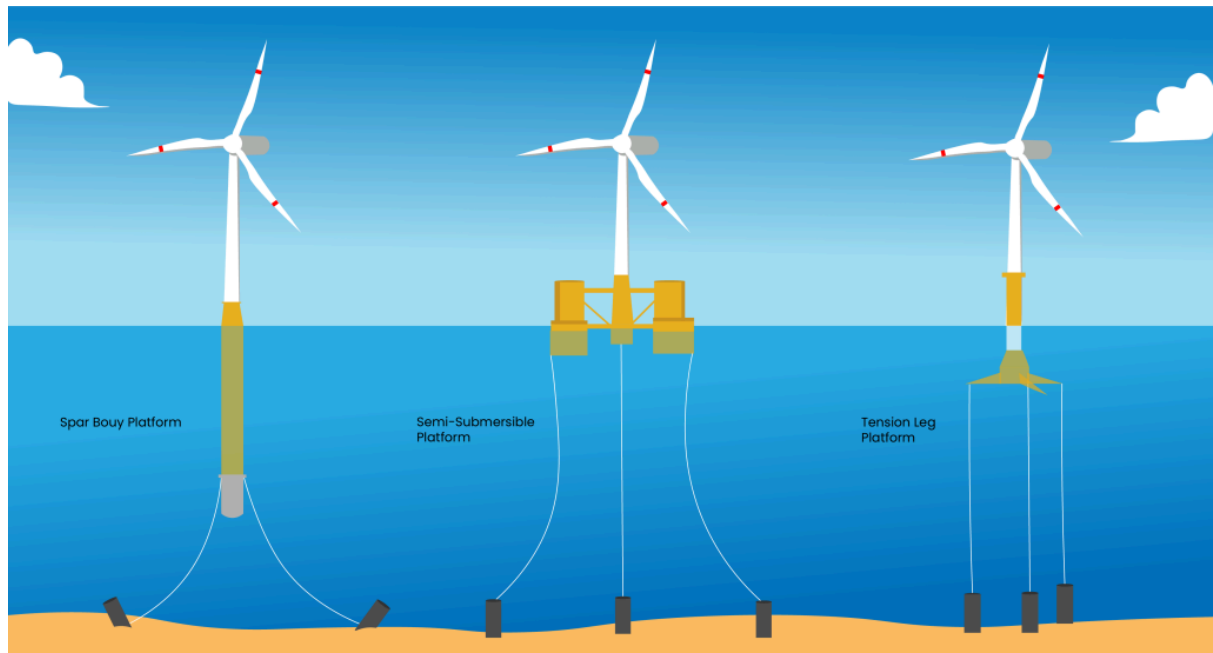
The proposed model achieved a reported **MAPE of 0.83%** and demonstrated effective anomaly detection using SCADA data. ([ScienceDirect](#))

Limitation

The method is designed specifically for wind turbines and requires suitable SCADA measurements.

Relevance

Feature selection and ensemble learning could improve the proposed system's **prediction accuracy and computational efficiency**.



Paper 5: An Embedded System for Remote Monitoring and Fault Diagnosis of Photovoltaic Arrays Using Machine Learning and IoT

Year: 2023

Research Problem

Solar PV systems need continuous monitoring to detect faults quickly. Cloud-only approaches may introduce latency and connectivity requirements.

Methodology

The researchers developed an embedded monitoring system combining:

- IoT
- Machine learning

- Edge/embedded computing
- Fault diagnosis

An artificial neural network was used for fault detection and the ML algorithms were integrated into a low-cost edge device. ([ScienceDirect](#))

Dataset/Technology

- PV-array measurements
- IoT sensors
- Machine learning
- Embedded processor

Major Findings

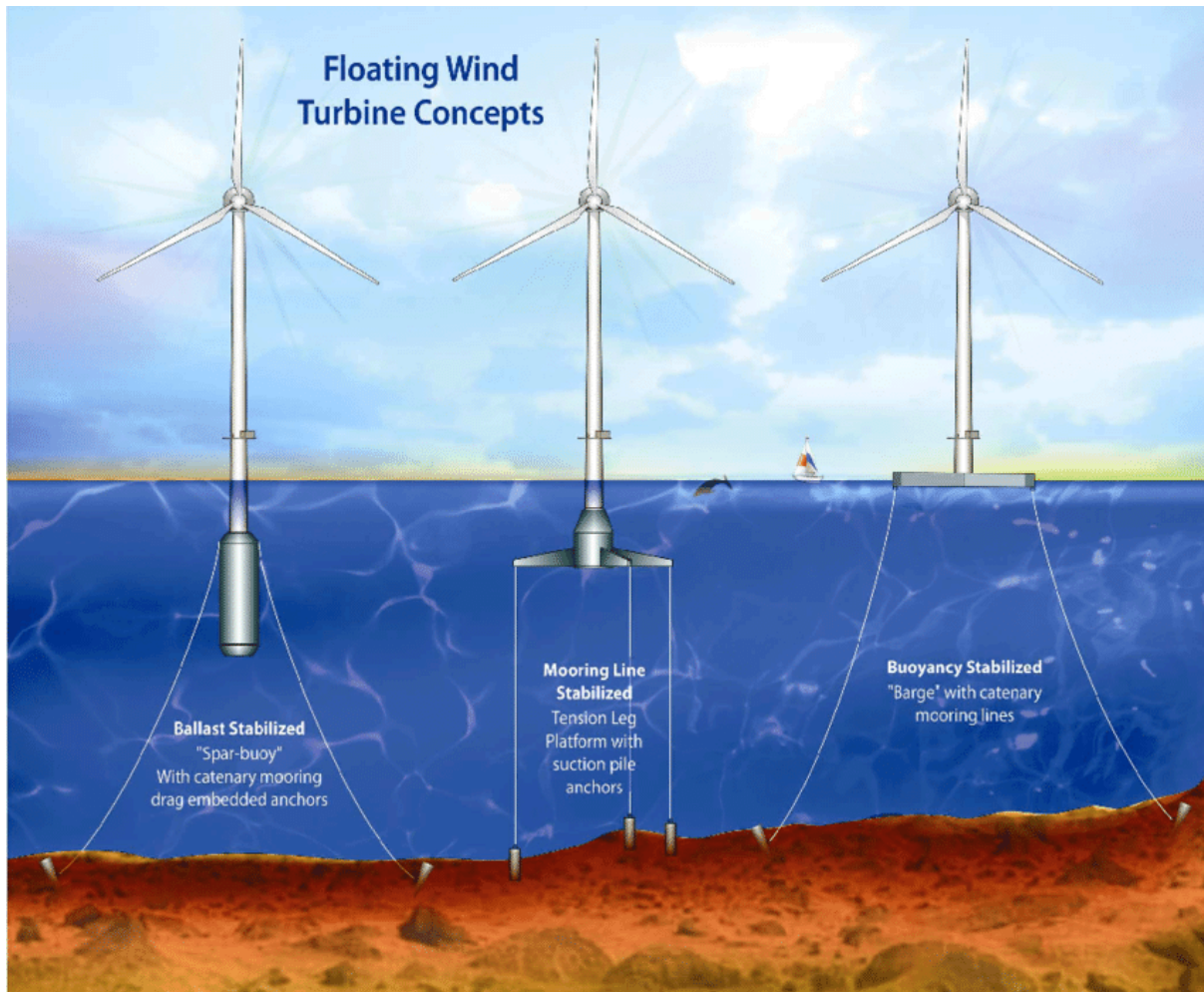
The research demonstrated the feasibility of performing PV fault diagnosis on an embedded platform while providing remote monitoring capabilities. ([ScienceDirect](#))

Limitation

Edge hardware has limited processing and storage resources compared with cloud platforms.

Relevance

This research is particularly important for the proposed system because it demonstrates **real-time monitoring close to the renewable-energy asset**.



Paper 6: Fault Detection in Solar Energy Systems: A Deep Learning Approach

Author/Year: Zeynep Bala Duranay – 2023

Research Problem

PV-panel defects can reduce energy generation, requiring fast and reliable detection.

Methodology

The study investigated deep-learning techniques using **infrared images of solar modules** for detecting PV-panel defects. ([MDPI](#))

Dataset/Technology

- Infrared solar-module images
- Deep learning
- Computer vision
- PV fault classification

Major Findings

The study demonstrated the potential of deep learning and infrared imaging for automated PV defect detection. ([MDPI](#))

Limitation

Image-based approaches may require cameras, suitable image quality, and additional computational resources.

Relevance

This approach could be integrated with **drone-based inspection** in the proposed monitoring platform.

Paper 7: Wind Turbine Fault Detection Based on SCADA Data Using Transformer Models

Authors/Year: Maldonado-Correa et al. – 2024

Research Problem

IGBT failures in wind-turbine converter systems can result in expensive downtime. Early prediction is therefore important.

Methodology

The researchers used Transformer-based deep-learning models with:

- SCADA operational data
- Alarm information
- Maintenance records
- Feature-selection techniques

The research was conducted using data from an onshore wind farm in Ecuador. ([ScienceDirect](#))

Major Findings

The models successfully predicted IGBT failures, with reported warnings approximately **4.25 months before failure**. ([ScienceDirect](#))

Limitation

The dataset was confidential, which limits reproducibility and independent comparison.

Relevance

Transformer models provide a promising approach for the proposed system's **long-term time-series failure prediction**.

Paper 8: Fault Detection in New Wind Turbines with Limited Data by Generative Transfer Learning

Year: 2025

Research Problem

New wind turbines often have insufficient historical fault data for training reliable AI models.

Methodology

The researchers proposed a **CycleGAN-based generative transfer-learning approach** to map SCADA data from data-rich turbines to data-scarce turbines.

Dataset

Field SCADA data from **seven substantially different wind turbines** was used to evaluate the approach. ([ScienceDirect](#))

Major Findings

The approach improved fault detection under limited-data conditions. Reported F1-score improvements were **10.3% with one month of training data and 16.8% with two weeks of data** compared with scarce-data normal behavior models. ([ScienceDirect](#))

Limitation

The approach is relatively complex and focuses on wind turbines.

Relevance

Transfer learning could help the proposed platform deploy AI models to **new renewable-energy plants without requiring years of historical data**.

4. Comparative Literature Survey Table

Author/ Year	Research Problem	Methodology	Dataset/ Tools	Key Findings	Limitations	Research Gaps
Chen et al. (2019)	Imbalanced wind-turbine fault data	Deep representation learning	Wind SCADA	Improved fault representation and generalization	Wind-focused	Need cross-asset monitoring
Xiang et al. (2021)	Early wind-turbine faults	CNN + LSTM + Attention	SCADA	Effective early-warning capability	Computational complexity	Need lightweight real-time deployment
Wind SCADA study (2022)	Wind-power prediction	RF, XGBoost, LSTM	Wind SCADA	XGBoost achieved $R^2 \approx 0.969$	Forecasting rather than maintenance	Integrating with predictive maintenance
Khan et al. (2023)	Wind-turbine anomalies	GA + XGBoost + RF + Extra Trees	SCADA	MAPE reported at 0.83%	Wind-specific	Need general renewable-asset model

Duranay (2023)	PV-panel defects	Deep learning + infrared imaging	PV infrared images	Automated defect detection	Image/data requirements	Combine image and sensor data
PV IoT study (2023)	PV remote monitoring	ML + IoT embedded system	PV-array data	Feasible edge-based monitoring	Hardware limitations	Hybrid edge-cloud architecture
Maldonado-Correa et al. (2024)	IGBT failure prediction	Transformer DL	SCADA + alarms + maintenance records	Failure warning ~4.25 months ahead	Confidential data	Need open and transferable models
Transfer-learning study (2025)	Limited wind-turbine data	CycleGAN transfer learning	7 wind turbines	Improved F1 under data scarcity	Complex implementation	Generalize across solar, wind and battery assets

5. Comparative Analysis

5.1 Machine Learning Approaches

Traditional ML algorithms such as **Random Forest** and **XGBoost** are effective for structured SCADA data. XGBoost showed strong

performance in wind-power forecasting, while ensemble approaches demonstrated effective anomaly detection. ([ScienceDirect](#))

5.2 Deep Learning Approaches

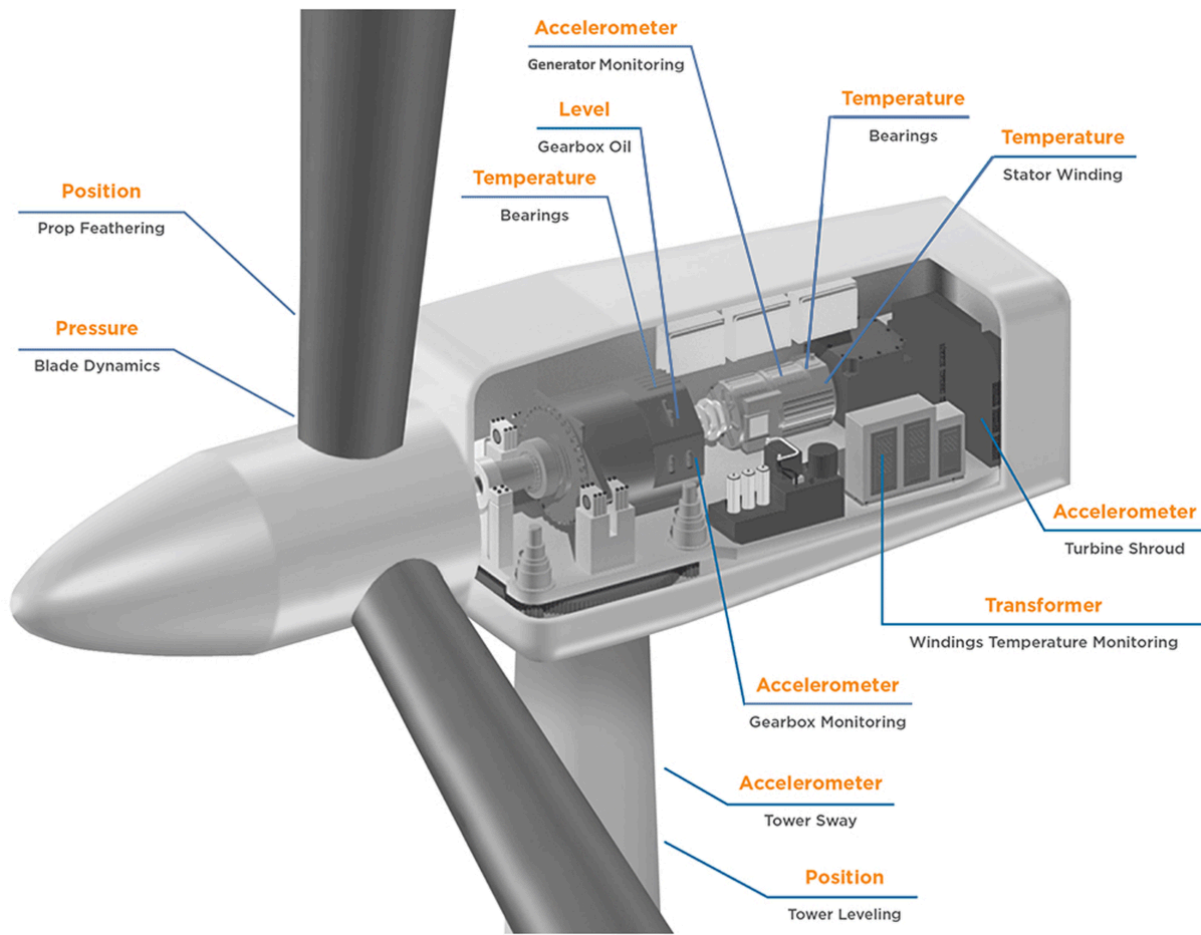
CNN-LSTM and Transformer models can capture complex temporal relationships in equipment data. CNN-LSTM with attention has demonstrated early fault-warning capability, while Transformer models have shown potential for long-term failure prediction. ([ScienceDirect](#))

5.3 Computer Vision

Deep learning applied to infrared or visual images can detect PV-panel defects. This provides an opportunity to incorporate **drone-based inspection** into renewable-energy monitoring. ([MDPI](#))

5.4 IoT and Edge Computing

IoT-based monitoring allows continuous collection of sensor information, while edge computing can perform rapid fault detection close to the equipment. The PV embedded-system research demonstrates the feasibility of this approach. ([ScienceDirect](#))

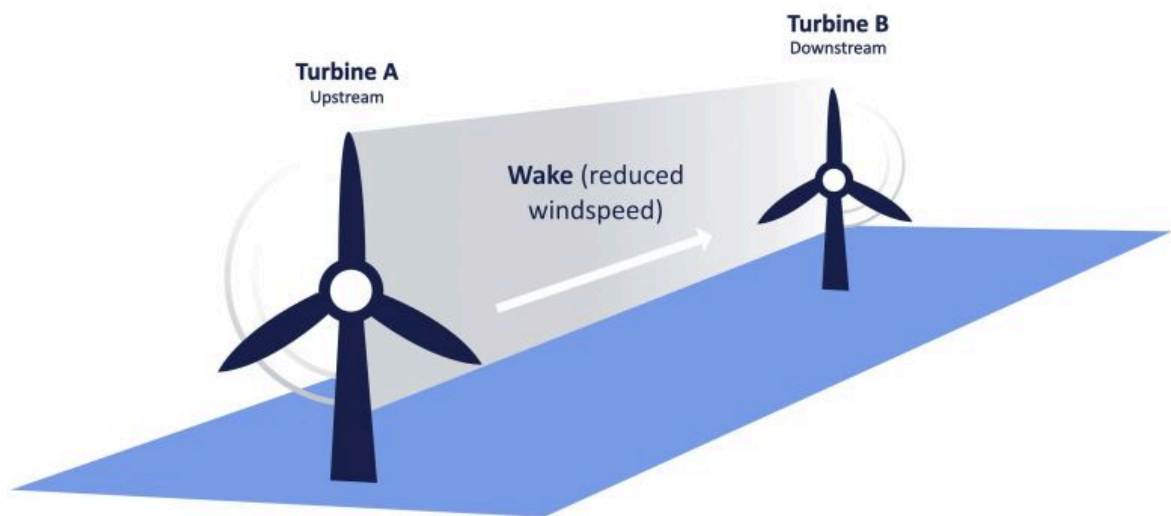


6. Limitations and Challenges Identified

The literature reveals several common challenges:

1. **Limited fault data** – Real equipment failures are relatively rare.
2. **Class imbalance** – Normal operating data is much larger than fault data.
3. **Data quality** – Missing, noisy, and inconsistent sensor measurements affect model accuracy.
4. **Limited generalization** – A model trained on one turbine or solar plant may not work equally well on another.
5. **Confidential datasets** – Many industrial studies cannot provide their datasets publicly.

6. **High computational requirements** – Deep-learning and Transformer models can be expensive to deploy.
 7. **Lack of explainability** – Operators may hesitate to trust predictions from black-box AI models.
 8. **Environmental variability** – Weather conditions can significantly affect renewable-energy performance.
 9. **Single-asset focus** – Many studies concentrate only on wind turbines or PV systems.
 10. **Real-time deployment challenges** – Research models are not always tested under actual continuous operational conditions.
-



7. Identified Research Gap

The major research gap identified from the literature is the **lack of a unified, intelligent, real-time monitoring platform capable of simultaneously monitoring different renewable-energy assets.**

Most existing studies focus on a particular problem:

- Wind-turbine fault detection
- PV-panel defect detection
- Wind-power forecasting
- Solar-power forecasting
- IoT-based PV monitoring

However, an actual renewable-energy plant may contain **solar panels, wind turbines, battery storage systems, weather stations, and grid-connected equipment.**

Therefore, there is a need for an integrated system that combines:

IoT Sensors → Real-Time Data → Cloud/Edge Processing → AI Prediction → Fault Detection → Weather Analysis → Maintenance Recommendation → Performance Dashboard

Another important gap is the limited ability of existing models to generalize across different sites and assets. Recent transfer-learning research shows that data scarcity remains an important challenge even within wind-turbine monitoring. ([ScienceDirect](#))

8. Proposed Improvements / Future Research Directions

The proposed system can address the identified gaps through the following improvements:

1. Multi-Asset Monitoring

Develop one platform for:

- Solar panels
- Wind turbines
- Battery energy storage
- Inverters
- Transformers

2. Hybrid AI Models

Combine:

- XGBoost/Random Forest for structured sensor data
- LSTM/Transformer for time-series data
- CNN/ViT for image-based inspection

3. Edge-Cloud Architecture

Use edge devices for immediate anomaly detection and cloud computing for large-scale analytics and model training.

4. Explainable AI

Use explainability techniques to show operators **why an asset has been classified as faulty**.

5. Transfer Learning

Transfer knowledge from existing plants to newly installed renewable assets, reducing the amount of historical data required.

6. Predictive Maintenance

Instead of simply detecting faults, the system should estimate:

- Failure probability
- Remaining useful life
- Maintenance priority
- Recommended maintenance date

- **7. Weather-Aware Prediction**

Combine weather forecasts with operational data to distinguish between **normal weather-related performance changes and actual equipment faults**.

8. Drone-Based Inspection

Use drone images and computer vision to identify:

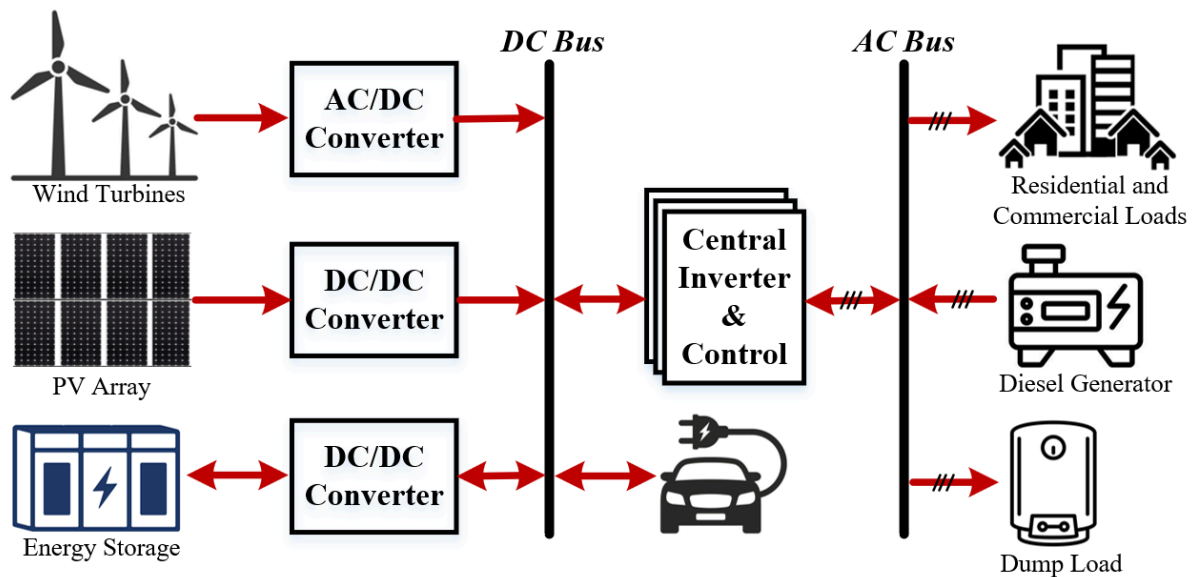
- Cracks
- Hotspots
- Dust
- Panel damage
- Surface defects

9. Real-Time Alerts

Provide notifications when abnormal performance or possible equipment failure is detected.

10. Continuous Model Updating

Implement online learning or periodic retraining so the AI model can adapt to changing environmental and equipment conditions.



9. Proposed Research Framework

The literature suggests the following architecture for the proposed system:

Renewable Energy Assets



IoT Sensors + SCADA + Drone Images + Weather Data



Edge Data Processing



Cloud Data Storage



Data Cleaning & Feature Engineering



AI/ML Models

- Fault Detection
- Failure Prediction
- Energy Forecasting
- Performance Analysis



Predictive Maintenance Engine

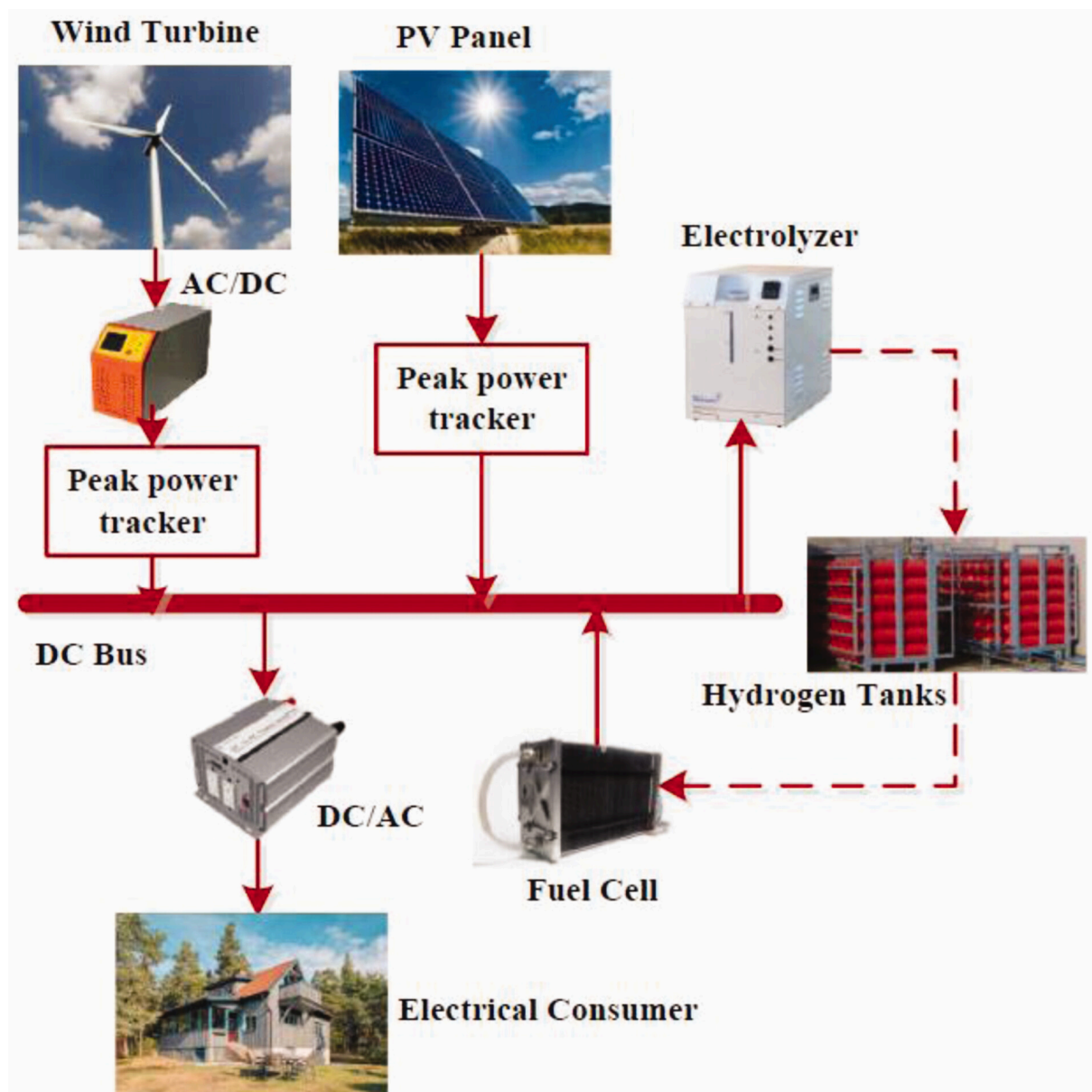


Recommendations & Alerts



Web/Mobile Monitoring Dashboard

This framework combines the strengths of the reviewed research while addressing the major gap of **integrated multi-asset renewable-energy monitoring**.



10. Conclusion

The literature survey shows that AI and machine learning have significant potential for improving renewable-energy asset management. SCADA-based ML models can identify abnormal turbine behavior, deep-learning models can detect complex temporal fault patterns, and computer-vision approaches can identify PV-panel defects. IoT and edge computing further enable real-time monitoring and faster response. ([ScienceDirect](#))

However, existing research is often limited to individual asset types, individual fault categories, or specific datasets. Data imbalance, limited failure records, model generalization, computational requirements, explainability, and real-world deployment remain important challenges.

Therefore, the proposed **AI-Based Smart Renewable Energy Asset Performance Monitoring System** can contribute by integrating IoT, SCADA, AI/ML, computer vision, weather analytics, edge/cloud computing, predictive maintenance, and performance reporting into a single platform. Such an integrated approach can help maximize energy generation, reduce downtime and maintenance costs, improve equipment reliability, and support sustainable renewable-energy production.

11. References — IEEE Style

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- [3] “SCADA system dataset exploration and machine learning based forecast for wind turbines,” *Results in Engineering*, vol. 16, 2022, Art. no. 100640. ([ScienceDirect](#))
- [4] P. W. Khan, C. Y. Yeun, and Y. C. Byun, “Fault detection of wind turbines using SCADA data and genetic algorithm-based ensemble learning,” *Engineering Failure Analysis*, vol. 148, 2023, Art. no. 107209. ([ScienceDirect](#))
- [5] Z. B. Duranay, “Fault Detection in Solar Energy Systems: A Deep Learning Approach,” *Electronics*, vol. 12, no. 21, 2023, Art. no. 4397. ([MDPI](#))
- [6] “An embedded system for remote monitoring and fault diagnosis of photovoltaic arrays using machine learning and the internet of things,” *Renewable Energy*, vol. 208, pp. 399–408, 2023. ([ScienceDirect](#))
- [7] J. Maldonado-Correa, J. Torres-Cabrera, S. Martín-Martínez, E. Artigao, and E. Gómez-Lázaro, “Wind turbine fault detection based on the transformer model using SCADA data,” *Engineering Failure Analysis*, vol. 162, 2024, Art. no. 108354. ([ScienceDirect](#))
- [8] “Fault detection in new wind turbines with limited data by generative transfer learning,” *Energy and AI*, vol. 22, 2025, Art. no. 100626. ([ScienceDirect](#))
- [9] “Photovoltaic power forecasting: A Transformer based framework,” *Energy and AI*, vol. 18, 2024, Art. no. 100444. ([ScienceDirect](#))
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